

Task 7 Report: Smart 5V Power Supply PCB Design using eSim

Submitted for: eSim Semester Long Internship - Autumn 2026

Tool Version: eSim v2.3 / v2.5 (KiCad-based)

1. Project Overview & Circuit Description

This project demonstrates the complete end-to-end design flow of a regulated **Smart 5V Power Supply PCB** created using eSim and KiCad. The board steps down and regulates an incoming DC voltage to deliver a clean, stable +5V output for electronic circuits.

Functional Circuit Breakdown:

- **Input Connector (J_1):** Provides the physical terminal interface for the incoming DC power supply.
- **Overcurrent Protection (F_1):** A fast-acting inline fuse that breaks the circuit if excessive current flows through the system.
- **Reverse Polarity Protection (D_1):** A series diode that prevents circuit damage if the input terminals are connected in reverse polarity.
- **Voltage Regulation (U_1 - L7805):** Linear voltage regulator that converts variable higher DC input into a constant +5V DC output.
- **Filtering & Noise Suppression (C_1, C_2):**
 - C_1 ($0.33\mu\text{F}$) stabilizes input ripple current near the regulator input pin.
 - C_2 ($0.1\mu\text{F}$) filters out high-frequency switching noise on the regulated output line.
- **Power Indicator (D_2 & R_1):** A 5mm LED (D_2) paired with a $1k\Omega$ current-limiting resistor (R_1) to provide immediate visual confirmation of active power.
- **Output Connector (J_2):** Terminal connector delivering the clean +5V output to downstream components.

2. Footprint Assignment Details

All component footprints were assigned using KiCad's footprint assignment tool during the design flow:

Referen ce	Value / Part	KiCad Footprint Library & Name
C_1	0.33	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm

C_2	0.1	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
D_1	Diode	Diode_THT:D_DO-41_SOD81_P10.16mm_Horizontal
D_2	LED	LED_THT:LED_D5.0mm
F_1	Fuse	Fuse:Fuseholder_Cylinder-5x20mm_Inline
J_1	2-pin Connector	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical
J_2	2-pin Connector	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical
R_1	$1k\Omega$	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
U_1	L7805	Package_TO_SOT_THT:TO-220-3_Vertical

3. Schematic Capture & PCB Design Flow

3.1 Schematic Capture

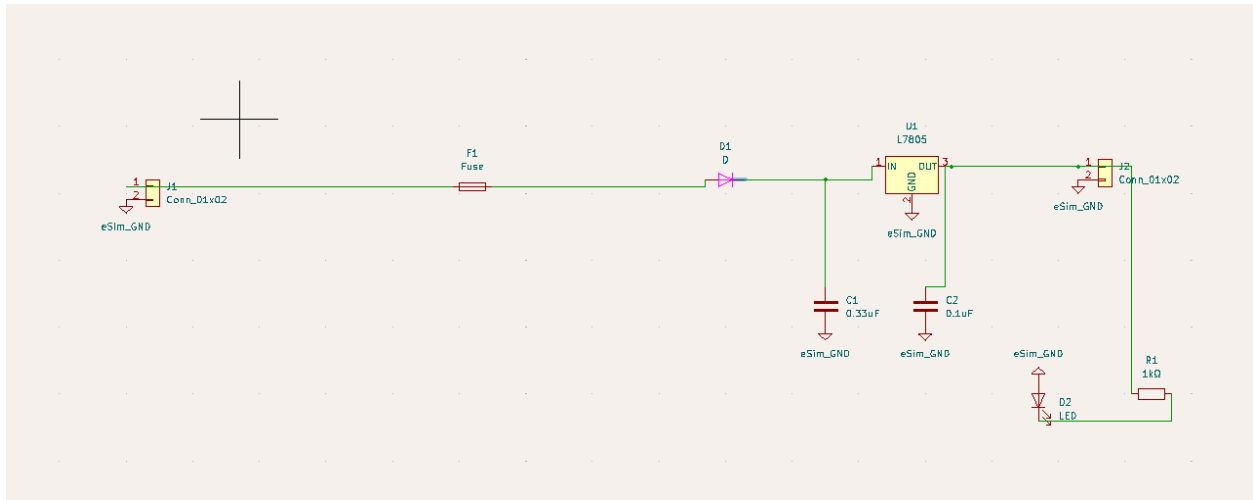


Figure 1: Complete Circuit Schematic in eSchema / KiCad Schematic Editor

3.2 Routed PCB Layout

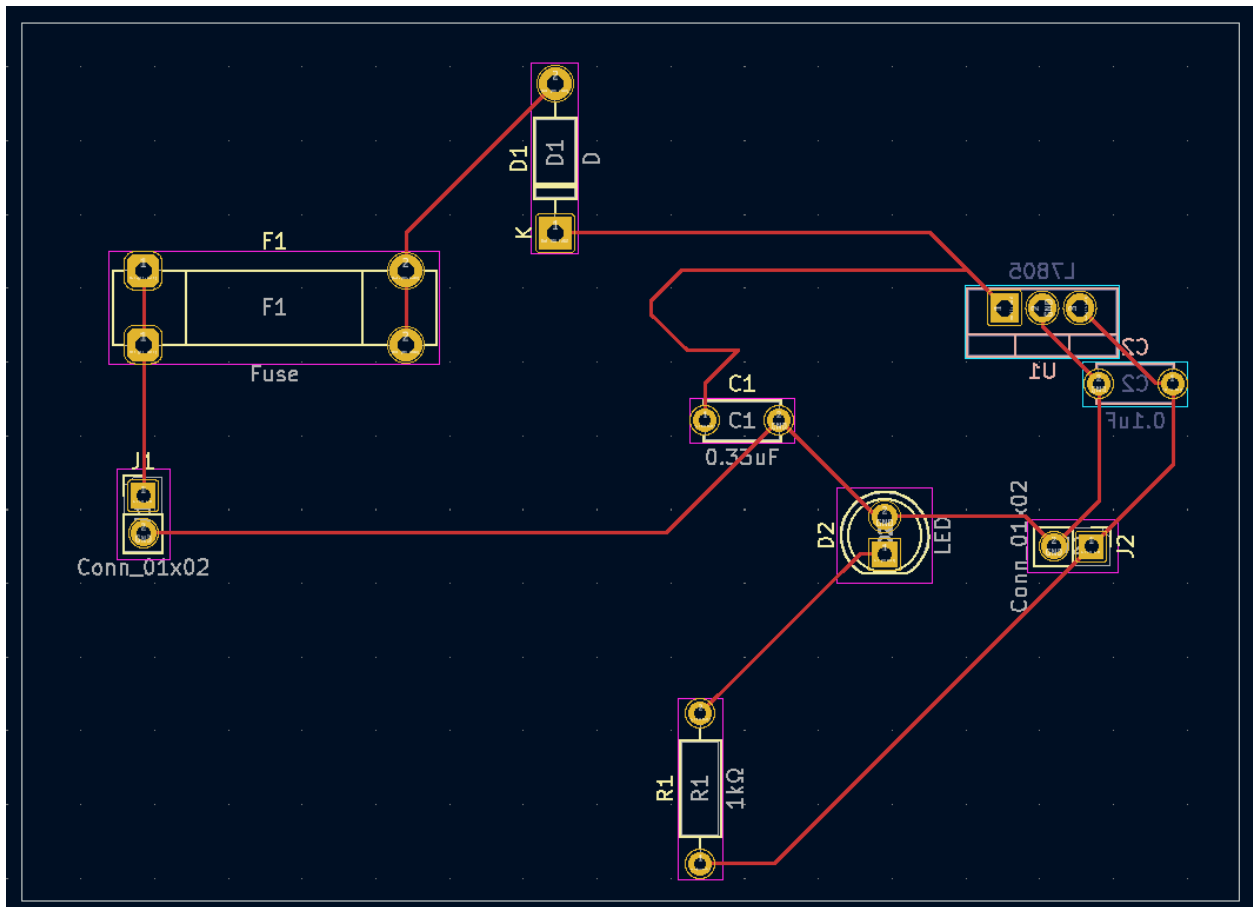


Figure 2: Fully Routed 2-Layer PCB Layout showing F.Cu copper tracks and Edge.Cuts boundary

3.3 Design Rule Check (DRC) Validation

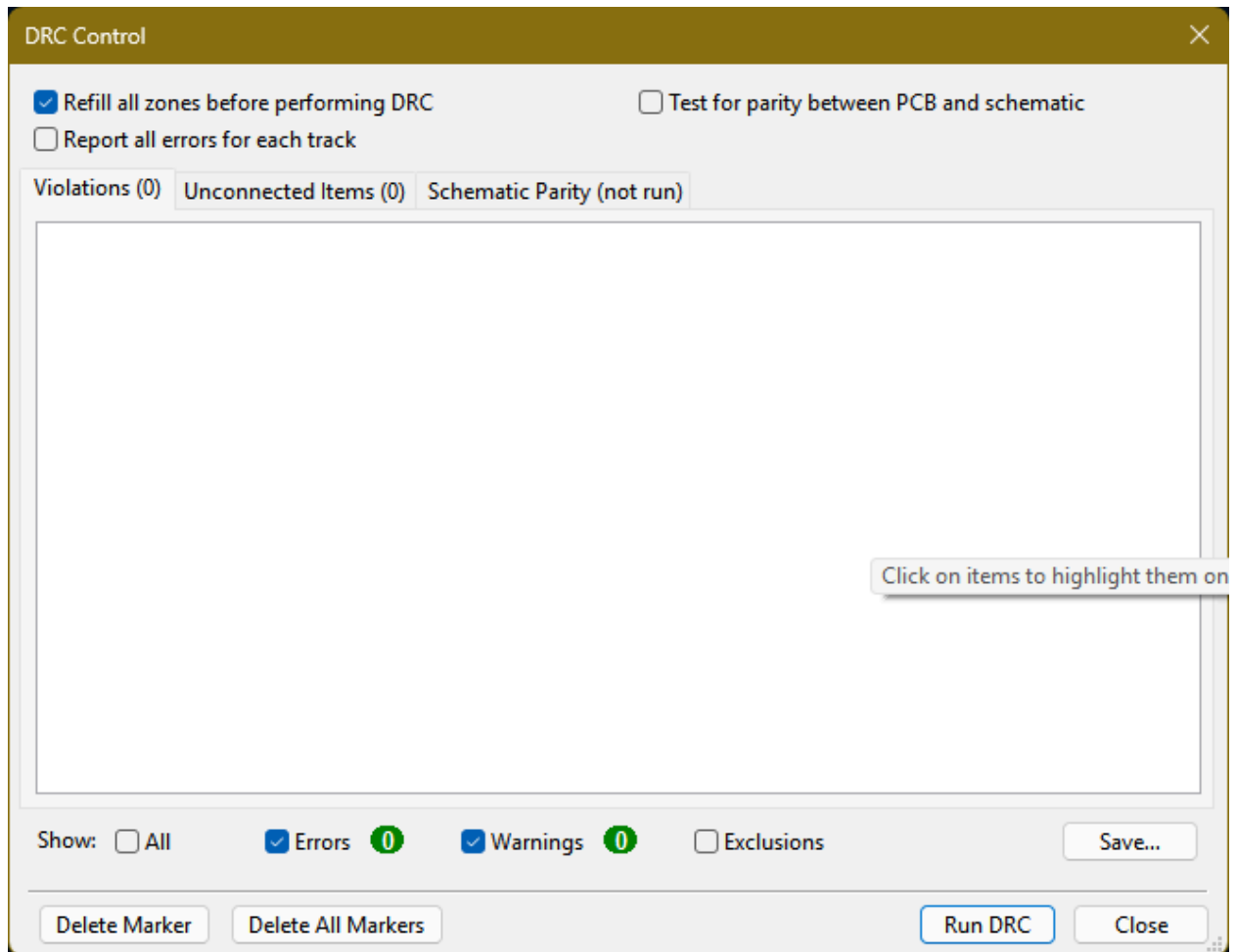


Figure 3: Design Rule Check (DRC) window confirming 0 Errors and 0 Warnings

3.4 3D Visualization

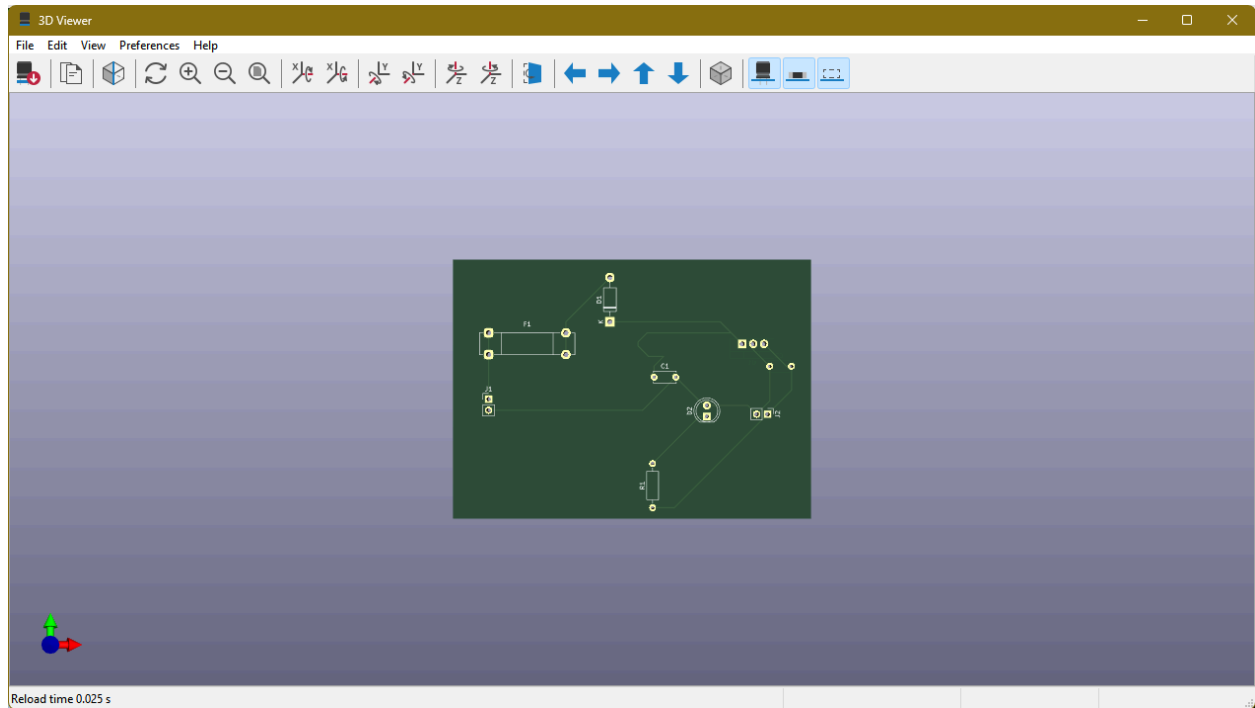


Figure 4: 3D Rendered view of the populated Smart 5V Power Supply PCB

4. Conclusion

The PCB layout strictly adheres to 2-layer design practices, incorporating proper trace clearances, protected input polarity, decoupling capacitor placement close to IC pins, and zero DRC errors. The eSim workspace opens seamlessly without library missing warnings.